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# Multiplicity Control in Oncology Clinical Trials With a Binary Surrogate Endpoint-Based Drop-The-Losers Design

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## ABSTRACT

Typical phase 1 oncology studies identify the maximum tolerated dose as the “optimal” dose for subsequent phases. With the advancement of molecular targeted agents and immunotherapies, however, evaluating two or more doses has become increasingly critical for dose selection. Such evaluation is often done in phase 2 studies in a randomized manner. In this article, we evaluate the strategy of applying an adaptive phase 2/3 seamless design for dose selection in oncology studies. Specifically, we consider the “drop-the-losers” design, where multiple treatment arms and a control arm are administered during the initial stage, and a more effective arm is identified for later stages by a binary surrogate endpoint such as overall response. We derive the theoretical type I error inflation scale and conduct simulation studies to illustrate the impact of various factors on the type I error inflation in such designs. Furthermore, we demonstrate the findings through the design of a lung cancer trial and introduce a software that implements the proposed design.

## 1 | Introduction

In traditional oncology drug development, dose-finding is primarily conducted in phase 1 clinical studies, where the maximum tolerated dose (MTD) is identified as the “optimal” dose for subsequent phases. However, with the increasing use of new molecular targeted agents and immunotherapies, the importance of dose-finding trials involving two or more doses in a randomized manner has become paramount, as highlighted in the Food and Drug Administration’s Project Optimus [1]. Consequently, the objective of phase 2 oncology clinical trials has been expanded to include the selection of optimal doses with respect for both toxicity and efficacy.

A significant body of research exists on dose selection in phase 2 trials. Much of this research pertains to multiple test procedures [2–4]. Furthermore, there is a growing interest in adaptive seamless multi-arm multi-stage phase 2/3 trials, because such approaches can reduce cost and time and accelerate the overall trial process. This includes designs based on combination tests [5, 6], Bayesian designs [7–9], designs that consider both efficacy and safety [10] or focus on both promising and compelling efficacy [11].

The “drop-the-losers” (DTL) design is a unique variant of the adaptive phase 2/3 seamless designs. In this approach, multiple treatment arms, along with a control arm, are administered during the initial stage. The most effective treatment arm, as determined by pre-specified selection criteria, is then chosen for further stages. Upon completion of the study, the final analysis is conducted to compare the selected treatment arm with the control arm, using data gathered throughout the study. The

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DTL design was initially proposed for continuous endpoints and two-stage trials [12]. It has since been expanded to include binomial endpoints, time-to-event endpoints, and multi-stage trials [13–15].

In DTL designs, type I error inflation is a major statistical concern [14]. This inflation arises from the correlation between the test statistics at the DTL and the final analysis stages. When the endpoints for both analyses are identical, assessing the correlation and, subsequently, the type I error inflation is relatively straightforward [16]. However, in oncology studies, the preferred approach is to select the arm based on intermediate or surrogate endpoints, such as overall response (OR) or pathological complete response (pCR), while the final analysis is conducted based on longer-term, time-to-event endpoints such as progression-free survival (PFS) or overall survival (OS) [17]. The combination test integrates $p$ values from different stages and controls the type I error in a strong sense [18, 19]; thus, can address the type I error inflation issue in DTL designs. However, the combination test does not account for the correlation between the surrogate and long-term endpoints, indicating potential enhancements to improve efficiency.

In this article, we focus on evaluating the correlation between binary surrogate endpoints, such as OR or pCR, and PFS or OS in oncology trials. We then examine the impact of this correlation on type I error inflation in an oncology trial with a DTL design. Specifically, we derive and validate a theoretical expression for this correlation and conduct simulation studies to illustrate the type I error inflation issue. Additionally, we assess the performance of our proposed solution for addressing this issue in various DTL scenarios.

The rest of the article is organized as follows. In Section 2, we delve into the DTL designs and the related type I error inflation issue. Specifically, we present the theoretical formulation of the type I error rate as well as the correlation coefficient between a binary OR or pCR endpoint and a time-to-event PFS or OS endpoint. In Section 3, we conduct simulation studies to validate our theoretical findings and assess the magnitude of type I error inflation in various scenarios. An example based on a real oncology trial is provided in Section 4. The DTL design R package is succinctly described in Section 5. Further discussion and considerations are offered in Section 6.

## 2 | Method

### 2.1 | The Drop-The-Losers (DTL) Design

Consider a randomized study that includes a control arm and $J$ treatment arms. Without loss of generality, we assume a 1:1 randomization ratio between any two arms that are enrolling. At a predetermined interim stage, referred to hereafter as the DTL look, one of the treatment arms is selected based on the observed data and is continued until the end of the study. Further analyses are conducted on the selected arm at the information fractions $t_1 < t_2 < \cdots < t_K = 1$ to compare it with the control arm with respect to the primary endpoint. Note that these information fractions are defined in relation to the selected arm and the control arm concerning the endpoint in the primary analysis at the end of the study. Figure 1 illustrates the overall scheme considered in this article.

Let $W_j$ be a statistic at the DTL look that corresponds to arm $j$, $j = 1, \ldots, J$. Based on the design, $W_j$ can be a test statistic comparing the treatment arm $j$ versus the control arm, or a summary statistic from arm $j$ only. Let $\boldsymbol{W} = (W_1, \ldots, W_J)$, and $g(\boldsymbol{W})$ be a arm-selection function with $g(\boldsymbol{W}) \in \{1, \ldots, J\}$. For example, $g(\boldsymbol{W}) = \text{argmax}_j W_j$, which means that the arm with the largest statistic $W_j$ at the DTL look will be selected.

Additionally, suppose treatment arm $j$ is selected at the DTL look. Then, let $H_0^{(j)}$ denote the primary hypothesis to be tested at the final analysis with respect to survival, such as PFS or OS:

$$H_0^{(j)} : \text{The survival functions of arm } j \text{ and the control are identical}$$

Let $Z_{j1}, \ldots, Z_{jK}$ be the test statistics for $H_0^{(j)}$ when comparing treatment arm $j$ versus control arm at information fractions

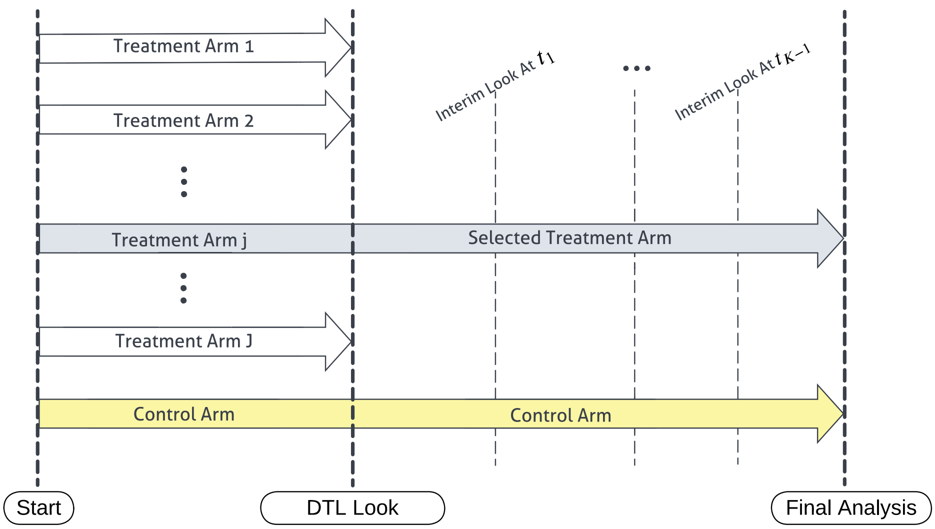


**FIGURE 1** | Overall scheme of DTL designs.



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$t_1, \ldots, t_K$, respectively, and let $c_{j1}, \ldots, c_{jK}$ be the corresponding group sequential design boundaries. Without loss of generality, a higher value of $Z_{jk}$ indicates a better treatment arm compared to the control arm. The primary hypothesis will be rejected if $Z_{j1} > c_{j1}, \ldots,$ or $Z_{jK} > c_{jK}$.

In this paper, unless stated differently, we use pCR as the surrogate endpoint at the DTL look, and PFS or OS as the primary endpoint at the final analysis. That is, $\boldsymbol{W}$ is based on observed pCR rates, and $Z_{j1}, \ldots, Z_{jK}$ are log-rank test statistics that compares the control arm to the treatment arm $j$ based on observed PFS or OS.

## 2.2 | Type I Error Inflation

Consider the hypothesis $H_0^{(j)}$ at the final analysis. Assume that the group sequential boundaries $c_{j1}, \ldots, c_{jK}$ control the type I error rate at $\alpha_j^*$ level. That is,

$$\Pr(\Pi_j) \le \alpha_j^*$$

where

$$\Pi_j = \{Z_{j1} > c_{j1}\} \cup \{Z_{j1} \le c_{j1}, Z_{j2} > c_{j2}\} \cup \cdots \cup \{Z_{j1} \le c_{j1}, \ldots, Z_{j,K-1} \le c_{j,K-1}, Z_{jK} > c_{jK}\}$$

Then, the rejection rate for $H_0^{(j)}$ is

$$r_j = \Pr(g(\boldsymbol{W}) = j, \Pi_j)$$

The overall rejection rate is then calculated as $r = \sum_{j=1}^{J} r_j$. Hereafter, without loss of generality, we will assume that $\alpha_j^* \equiv \alpha^*$, and $c_{j1} \equiv c_1, \ldots, c_{jK} \equiv c_K$ for all $j$.

When $W_j \perp\!\!\!\perp (Z_{j1}, \ldots, Z_{jK})\ \forall\ j$,

$$r = \sum_{j=1}^{J} \Pr(g(\boldsymbol{W}) = j, \Pi_j) = \sum_{j=1}^{J} \Pr(g(\boldsymbol{W}) = j) \Pr(\Pi_j) \le \alpha^* \sum_{j=1}^{J} \Pr(g(\boldsymbol{W}) = j) = \alpha^*$$

This implies that the family-wise error rate (FWER) is controlled at the $\alpha^*$ level. However, the independence assumption will not hold in typical clinical trial scenarios, and the FWER could markedly exceed $\alpha^*$. Examples illustrating inflation are provided below.

**Example 1.** Consider $J = 2$ and $K = 1$. For this example, suppose PFS is the endpoint both at the DTL look and the final analysis. In this setting, the primary hypothesis $H_0^{(j)}$ is that there is no difference in PFS between arm $j$ and control. Let $Z_{j1}$ be the log-rank test statistic for $H_0^{(j)}$. Denote $t_0$ as the information fractionand $Z_{j0}$ be the log-rank test statistic for $H_0^{(j)}$at the DTL look. Let $W_1 = Z_{10}$ and $W_2 = Z_{20}$, and specify the arm-selection function

$$g(W_1 = Z_{10}, W_2 = Z_{20}) = \mathrm{I}(Z_{20} - Z_{10} - \Delta \le 0) + 2\mathrm{I}(Z_{20} - Z_{10} - \Delta > 0)$$

Here, $\Delta$ represents a predefined threshold, and the arm-selection function $g(\ )$ specifies that if $Z_{20} - Z_{10} > \Delta$, then arm 2 will be selected; otherwise, arm 1 will be selected.

Under the global null hypothesis, $H_0 = H_0^{(1)} \cap H_0^{(2)}$, there is no difference in PFS among treatment arms 1 and 2 and the control arm. In this case, the joint distribution of the log-rank test statistics $(Z_{10}, Z_{20}, Z_{11}, Z_{21})$ is a multivariate normal distribution [16] with mean $\mathbf{0}$ and covariance

$$\Sigma = \begin{bmatrix} 1 & \frac{1}{2} & \sqrt{\frac{t_0}{t_1}} & \frac{1}{2}\sqrt{\frac{t_0}{t_1}} \\ \frac{1}{2} & 1 & \frac{1}{2}\sqrt{\frac{t_0}{t_1}} & \sqrt{\frac{t_0}{t_1}} \\ \sqrt{\frac{t_0}{t_1}} & \frac{1}{2}\sqrt{\frac{t_0}{t_1}} & 1 & \frac{1}{2} \\ \frac{1}{2}\sqrt{\frac{t_0}{t_1}} & \sqrt{\frac{t_0}{t_1}} & \frac{1}{2} & 1 \end{bmatrix}$$

In general, all log-rank test statistics marginally follow a standard normal distribution, and the correlations depend on the common patients from treatment or control arms.

Let $z_{\alpha^*}$ denote the $(1-\alpha^*)$ percentile of the standard normal distribution. If $H_0^{(1)}$ is true and $H_0^{(2)}$ is false, the FWER can be represented as:

$$r = r_1 = \Pr(Z_{20} - Z_{10} - \Delta \le 0, Z_{11} > z_{\alpha^*}) \le \Pr(Z_{11} > z_{\alpha^*}) = \alpha^*$$

Similarly, when $H_0^{(2)}$ is true and $H_0^{(1)}$ is false, the FWER is

$$r = r_2 = \Pr(Z_{20} - Z_{10} - \Delta > 0, Z_{21} > z_{\alpha^*}) \le \Pr(Z_{21} > z_{\alpha^*}) = \alpha^*$$

In both cases, the FWER is controlled at the $\alpha^*$ level.

Under the global null, $H_0 = H_0^{(1)} \cap H_0^{(2)}$, the FWER is given by:

$$r = r_1 + r_2 = \alpha^* + \int_{z_{\alpha^*}}^{\infty} \left[\Phi(d_1) - \Phi(d_2)\right] \phi(z) dz$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ represent the standard normal probability density function (PDF) and cumulative distribution function (CDF), respectively, and

$$d_1 = \frac{2\Delta + z\sqrt{t_0/t_1}}{\sqrt{4 - t_0/t_1}}, \quad d_2 = \frac{2\Delta - z\sqrt{t_0/t_1}}{\sqrt{4 - t_0/t_1}}$$

The derivation of $d_1$ and $d_2$ can be found in Appendix A. It is clear that $d_1 > d_2$, thus $r > \alpha^*$, leading to type I error inflation.

**Example 2.** Consider $J = 2$ and $K = 1$. The trial utilizes pCR as the endpoint at the DTL look, but PFS is the endpoint at the final analysis. Let $W_j$ be the observed pCR rate—the mean of the binary pCR indicator for all patients—in arm $j$ at the DTL look. The arm-selection function is the same as Example 1: $g(W_1, W_2) = \mathrm{I}(W_2 - W_1 - \Delta \le 0) + 2\mathrm{I}(W_2 - W_1 - \Delta > 0)$. Similarly, $\Delta$ is a predefined threshold, and the arm-selection function $g(\ )$ specifies that if the pCR rate in arm 2 is higher than that in arm 1 by more than $\Delta$, arm 2 is selected; otherwise, arm 1 is chosen.

Let $n$ be the sample size for each arm at the DTL look. Under the global null, let $q = E(W_1) = E(W_2)$, and let $\rho$ be the correlation



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coefficient of $W_j$ and $Z_{j1}$ for $j = 1, 2$. Without loss of generality, we let $\rho \geq 0$. Then,

$$(W_1, W_2, Z_{11}, Z_{21}) \sim \text{MVN}\left(\begin{bmatrix} q \\ q \\ 0 \\ 0 \end{bmatrix}\begin{bmatrix} \frac{q(1-q)}{n} & 0 & \rho\sqrt{\frac{q(1-q)}{n}} & 0 \\ 0 & \frac{q(1-q)}{n} & 0 & \rho\sqrt{\frac{q(1-q)}{n}} \\ \rho\sqrt{\frac{q(1-q)}{n}} & 0 & 1 & \frac{1}{2} \\ 0 & \rho\sqrt{\frac{q(1-q)}{n}} & \frac{1}{2} & 1 \end{bmatrix}\right) \quad (1)$$

Subsequently, we can derive the FWER

$$r = \alpha^* + \int_{z_{\alpha^*}}^{\infty} \left[\Phi(d_1) - \Phi(d_2)\right]\phi(z)dz \quad (2)$$

with

$$d_1 = \frac{\Delta/\sqrt{q(1-q)/n} + \rho z}{\sqrt{2-\rho^2}}, \; d_2 = \frac{\Delta/\sqrt{q(1-q)/n} - \rho z}{\sqrt{2-\rho^2}} \quad (3)$$

Details can be found in Appendix A. Moreover, the derivation of FWER can be expanded to scenarios with multiple interim analyses ($K > 0$) and multiple treatment arms ($J > 1$). We provide the details in the Supporting Information.

A few observations based on (2) and (3) can be made. First, when $\rho = 0$, $r = \alpha^*$, implying that there will be no type I error inflation when the independence assumption holds. Second, when $\Delta = 0$, as $\rho$ increases, the FWER increases. Intuitively, the FWER increases as $\rho$ increases for any $\Delta$; however, there is no theoretical proof to support this yet. Third, when $\Delta = 0$, the FWER is independent of $n$ and $q$. Fourth, $r$ monotonically increases as $\alpha^*$ increases, thus if we want to control the FWER at level $\alpha$, we can numerically obtain the unique $\alpha^*$ such that $r = \alpha$. Lastly, let $\Delta^\dagger = \Delta/\sqrt{q(1-q)/n}$. Then, $r \to \alpha^*$ as $\Delta^\dagger \to \infty$. However, the relationship between $r$ and $\Delta^\dagger$ isn't monotonic, as

$$\frac{\partial r}{\partial \Delta^\dagger} = \frac{1}{\sqrt{2-\rho^2}} \int_{z_{\alpha^*}}^{\infty} \left[\phi(d_1) - \phi(d_2)\right]\phi(z)dz$$

does not appear to be strictly positive or negative.

In Section 3.1, we conduct simulation studies to further illustrate the extent of type I error inflation under various scenarios for Example 2.

### 2.3 | Correlation Coefficient $\rho$

At the design stage, one may conservatively set $\rho = 1$ when selecting $\alpha^*$ to ensure $r \leq \alpha$ in Equation (2). This strategy was adopted by [20] when evaluating the FWER in a study with a group sequential design and multiple correlated endpoints. To better preserve power while controlling the FWER, however, a better understanding of $\rho$ will be necessary.

For patient $i$ in arm $j$ ($j = 0, \ldots, J$, where $j = 0$ represents the control arm), let $X_{ij}$ denote the binary pCR indicator and $T_{ij}$ denote the time to the PFS event. Let $n$ be the number of patients in each arm at the DTL look, and $n_k$ ($k = 1, \ldots, K$) be the number of patients in the selected and the control arms at the $k$th interim analysis. Let $\tau_k = n/n_k$. Furthermore, we consider $W_j = \frac{1}{n}\sum_{i=1}^{n} X_{ij}$ for $j = 1, \ldots, J$, and let $Z_{jk}$ represent the log-rank test statistics comparing the control arm and treatment arm $j$ at the $k$th interim (or final) analysis.

Under the global null assumption, let $T_{ij}|X_{ij} = x \sim f_x(t)$ for $x = 0, 1$. Let $S_x(t)$ be the survival function corresponding to $f_x(t)$ for $x = 0, 1$. Then, $T_{ij}$ has marginal density function $f(t) = f_1(t)q + f_0(t)(1-q)$, survival function $S(t) = S_1(t)q + S_0(t)(1-q)$, and hazard function $h(t) = f(t)/S(t)$.

**Proposition 1.** *Denote $\rho_{jk} = \text{Cor}(W_j, Z_{jk})$. Without censoring,*

$$\rho_{jk} \approx \sqrt{\frac{q\tau_k}{2(1-q)}}\left[\int_0^{\infty} \frac{S_1(t)f(t)}{S(t)}dt - 1\right] \quad (4)$$

*Under the proportional hazard assumption $S_1(t) = [S_0(t)]^\gamma$,*

$$\rho_{jk} \approx \sqrt{\frac{q\tau_k}{2(1-q)}}\left(\frac{1}{\gamma} - 1\right)E\left[\frac{1}{1 + \frac{q}{1-q}U^{1-\frac{1}{\gamma}}}\right] \quad (5)$$

*where $U \sim$ Uniform(0, 1).*

**Proposition 2.** *With non-informative censoring, where the censoring time follows a cumulative distribution function, $1 - S_C(t)$,*

$$\rho_{jk} \approx \frac{\sqrt{\frac{q\tau_k}{2(1-q)}}\int_0^{\infty}\left[\frac{S_1(t)f(t)}{S(t)} - f_1(t)\right]S_C(t)dt}{\sqrt{\int_0^{\infty} f(t)S_C(t)dt}} \quad (6)$$

*Under the proportional hazard assumption $S_1(t) = [S_0(t)]^\gamma$,*

$$\rho_{jk} \approx \frac{\sqrt{\frac{q\tau_k}{2(1-q)}}\left(\frac{1}{\gamma} - 1\right)\int_0^{\infty} \frac{1}{1+\frac{q}{1-q}[S_1(t)]^{1-\frac{1}{\gamma}}} f_1(t)S_C(t)dt}{\sqrt{\int_0^{\infty} f(t)S_C(t)dt}} \quad (7)$$

**Theorem 1.** *Under the proportional hazard assumption $S_1(t) = [S_0(t)]^\gamma$ with$\gamma \leq 1$, $\forall S_C(t)$,*

$$\rho_{jk} \leq \sqrt{\frac{q\tau_k}{2(1-q)}}\left(\frac{1}{\gamma} - 1\right)\sqrt{E\left[\frac{1}{\left(1 + \frac{q}{1-q}U^{1-\frac{1}{\gamma}}\right)^2\left(q + \frac{1-q}{\gamma}U^{\frac{1}{\gamma}-1}\right)}\right]} \quad (8)$$

*where $U \sim$ Uniform(0, 1).*

We prove Propositions 1 and 2, and Theorem 1 in Appendices Proof of Propositions and Proof of Theorem. Furthermore, we conduct simulation studies in Section 3.2 to numerically evaluate the approximations and boundary.

### 2.4 | Type I Error Control

In summary, the DTL design considers the following parameters for type I error control: The number of patients in each arm at



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the DTL look, $n$; the response rate, $q$; the arm selection function, $g(\boldsymbol{W})$, or more specifically, the arm selection criteria, $\Delta$; the information fraction at the interim analyses, $\{t_k\}$; and the correlations between $W_j$ and $Z_{jk}$, $\{\rho_{jk}\}$.

Given these design parameters $\Omega = \{n, q, \Delta, \{t_k\}, \{\rho_{jk}\}\}$, the task of type I error control becomes identifying

$$\widetilde{\alpha} = \sup_{\alpha^*}\{\alpha^* : r(\Omega) \leq \alpha\} \tag{9}$$

where $\alpha^*$ is the type I error rate based on which the group sequential design boundary is derived.

In practice, we propose to conservatively identify

$$\widetilde{\alpha} = \sup_{\alpha^*}\left\{\alpha^* : \left\{\sup_{q\in Q_q,\gamma\in Q_\gamma,\{\rho_{jk}=\rho^*_{jk}(\tau_k,q,\gamma)\}} r(\Omega)\right\} \leq \alpha\right\} \tag{10}$$

where $Q_q$ and $Q_\gamma$ are the parameter spaces of $q$ and $\gamma$ that are clinically meaningful, respectively, and $\rho^*_{jk}(\tau_k, q, \gamma)$ is the boundary for $\rho_{jk}$ in Equation (8), given $\tau_k$, $q$ and $\gamma$.

In Section 4, we illustrate the identification of $\widetilde{\alpha}$ through the design of a real oncology study.

## 3 | Simulation Study

### 3.1 | Type I Error Inflation

In this section, we conduct simulation studies to numerically assess the type I error inflation in various DTL scenarios.

The data generation setting follows Example 2 in Section 2.2 and reflects the data observed at the final analysis. Specifically, we consider the correlation between test statistics of pCR status and PFS, $\rho \in \{0, 0.1, \ldots, 0.7\}$; the pCR rate, $q \in \{0.3, 0.5, 0.95\}$; and the sample size of each arm at the DTL look, $n \in \{50, 150, 300\}$. Furthermore, we consider $g(W_1, W_2) = \mathrm{I}(W_2 - W_1 - \Delta \leq 0) + 2\mathrm{I}(W_2 - W_1 - \Delta > 0)$ and $\Delta \in \{0, 0.1\}$. Lastly, we set $\alpha = 0.025$. Based on 1 000 000 replications, selected results are presented in Figure 2 and Table 1. Comprehensive results are available in the Supporting Information.

In the absence of type I error control for the DTL look, Figure 2 demonstrates that the type I error can be inflated to 0.045 when $\rho = 0.7$ and $\Delta = 0$. The severity of this inflation is a monotonically increasing function of $\rho$ alone when $\Delta = 0$. When $\Delta \neq 0$, the type I error inflation is also influenced by the sample size at the DTL look $n$ and the pCR rate $q$. As $n$ increases or $q \to 1$, the FWER converges to 0.025. These findings are consistent with the results discussed in Section 2.2. Additionally, Figure 2 indicates that the simulation results align with the theoretical results.

When using $\tilde{\alpha}$ as defined in Equation (9) for the analysis following the DTL look, we report in the Table 1 that the FWER is well controlled. The complete version of Table 1 is available in the Supporting Information.

### 3.2 | Evaluating the Magnitude and Impact of $\rho$

In this section, we conduct simulation studies to assess the magnitude of $\rho$ under different settings. Additionally, we evaluate the approximation and the boundary in Section 2.3.

The data generation follows the same scenario as Example 2 in Section 2.2. We consider the sample size of each arm at the DTL look, $n \in \{50, 150, 300\}$, and the total sample size of the selected arm $N = n/\tau$, with $\tau \in \{1/3, 2/3\}$. For PFS, we assume $f_0(t) = \text{Exponential}(\lambda)$ for non-responders and $f_1(t) = \text{Exponential}(\gamma\lambda)$ for responders, with $\lambda \in [1.6, 2.6]$ and $\gamma \in [0.1, 0.9]$. In addition to the scenario with no censoring, we examine censoring times following two different uniform distributions: Uniform$(0, 2.2)$, referred to as moderate censoring, and Uniform$(0, 6.5)$, referred to as minimal censoring. These distributions result in censoring rates ranging from 7% to 42%.

Based on 100 000 replications, results are presented in Figure 3. Comprehensive results are available in the Supporting Information. Several key observations can be made. First, given $\tau$, $\rho$ does not depend on $n$, which is advantageous for study design. Second, as $\tau$ increases, $\rho$ also increases because the amount of information at the DTL look relatively increases. Third, for $\rho$ versus $q$ in the left panel, $\rho \to 0$ as $q \to 0$ or 1, and the relationship is quadratic. Fourth, for $\rho$ versus $\gamma$ in the center panel, $\rho$ increases as the hazard ratio $\gamma$ between responders and non-responders increases, although we lack theoretical proof that the relationship is monotonic. Notably, $\rho$ does not monotonically increase or decrease as the amount of censoring increases. For instance, for $\gamma = 0.1$ and $\tau = 2/3$, $\rho = 0.38, 0.40, 0.38$ for moderate censoring, minimal censoring, and no censoring, respectively. Lastly, for $\lambda$ versus $\rho$ in the right panel, the magnitude of $\rho$ is minimally affected by $\lambda$.

In Figure 3, we present the simulation results against the theoretical approximations and upper bounds as defined in Equations (5), (7), and (8). The approximations demonstrate congruence with the simulation outcomes, with all $\rho$ values consistently falling within the theoretical upper bounds. Notably, the upper bound of $\rho$ does not exceed 0.7 across all scenarios of $\tau$, $q$ and $\gamma$ considered.

We also conduct simulation studies in which the PFS follows a Weibull distribution. The results align closely with those observed in the exponential case, demonstrating that the theoretical approximations remain robust under different distribution assumptions. Details are provided in the Supporting Information.

## 4 | Application

In this section, we apply the DTL design to a hypothetical PD-1 and LAG-3 combination therapy study for patients with non-small cell lung cancer (NSCLC). We consider two treatment arms, High Dose (HD) and Low Dose (LD), and compare them to the standard of care (SOC).



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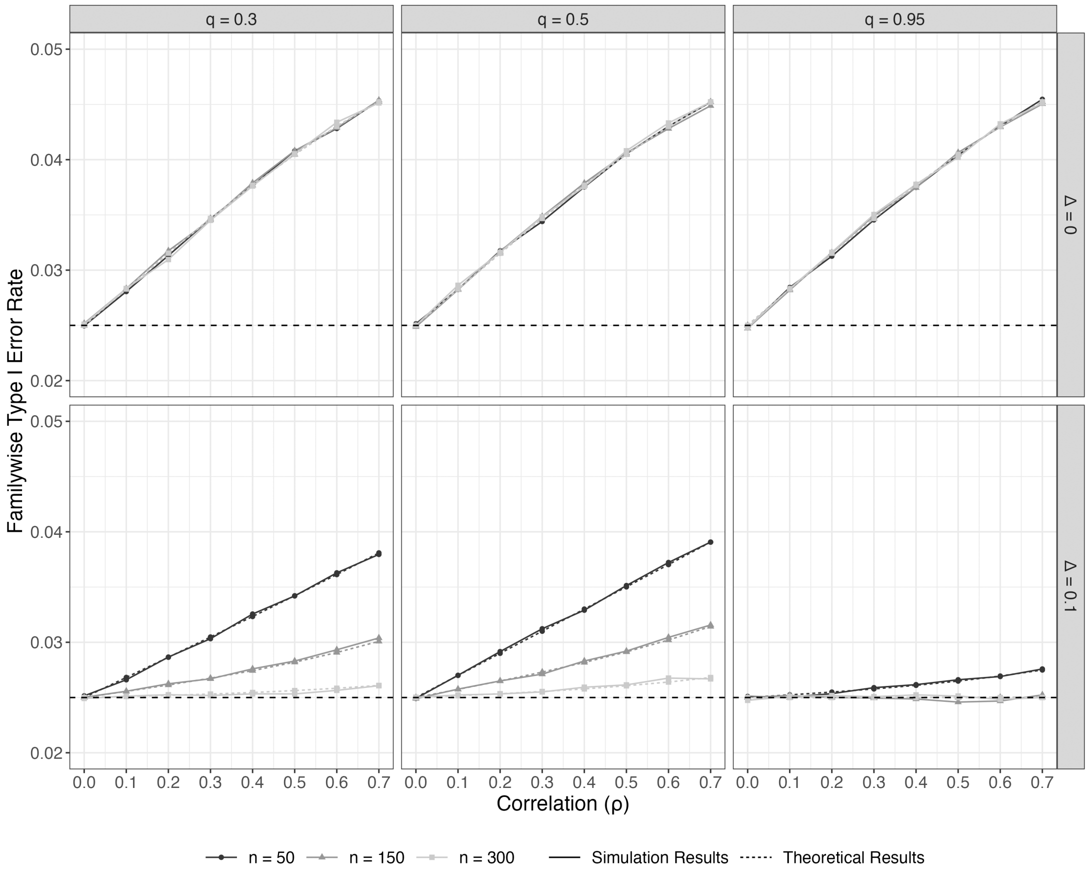


**FIGURE 2** | Empirical and theoretical FWERs in DTL designs without type I error control.

**TABLE 1** | Simulated FWERs using $\widetilde{\alpha}$ at the final analysis for $n = 150$ and $\alpha = 0.025$.

| | | $q = 0.30$ | | $q = 0.50$ | | $q = 0.95$ | |
|---|---|---|---|---|---|---|---|
| **Δ** | $\rho$ | $\widetilde{\alpha}$ | **FWER** | $\widetilde{\alpha}$ | **FWER** | $\widetilde{\alpha}$ | **FWER** |
| 0 | 0.1 | 0.0221 | 0.0250 | 0.0221 | 0.0250 | 0.0221 | 0.0253 |
| | 0.5 | 0.0151 | 0.0248 | 0.0151 | 0.0252 | 0.0151 | 0.0249 |
| | 0.7 | 0.0135 | 0.0249 | 0.0135 | 0.0248 | 0.0135 | 0.0248 |
| 0.1 | 0.1 | 0.0245 | 0.0251 | 0.0243 | 0.0252 | 0.0250 | 0.0249 |
| | 0.5 | 0.0221 | 0.0249 | 0.0213 | 0.0248 | 0.0250 | 0.0251 |
| | 0.7 | 0.0206 | 0.0252 | 0.0196 | 0.0251 | 0.0250 | 0.0250 |

## 4.1 | Initial Design

The initial power and sample size calculations are as follows: The median progression-free survival (mPFS) in the SOC arm is expected to be 180 days. Both the HD and LD arms are expected to have the same mPFS, with an expected hazard ratio of 0.6 versus SOC. The annual drop-out rate is anticipated to be 5%, and the monthly enrollment rate is projected to be 20 patients per arm. Consequently, the study will require $N = 152$ subjects per arm and a total of $D = 162$ events to achieve 90% power at a one-sided alpha level of 0.025 to detect a significant difference between HD and SOC, or between LD and SOC.

Based on clinical, regulatory, and operational considerations, the study plans to conduct a DTL analysis based on pCR when there are 80 patients in each arm (i.e., $n = 80$). If the pCR rate in the HD arm is more than 5% better than that in the LD arm (i.e., $\Delta = 0.05$), the study will drop the LD arm. Otherwise, the study will drop the HD arm.

After the DTL look, an interim analysis for PFS will be conducted at an information fraction of 0.5. The O'Brien–Fleming boundaries [21] will be used for the group sequential design.

## 4.2 | Type I Error Control

First, we quantify the correlation coefficient $\rho$ by analyzing the data from the PD-1 arm of an existing study. The 95% confidence interval of the pCR rate was $[0.19, 0.32]$, and the 95%



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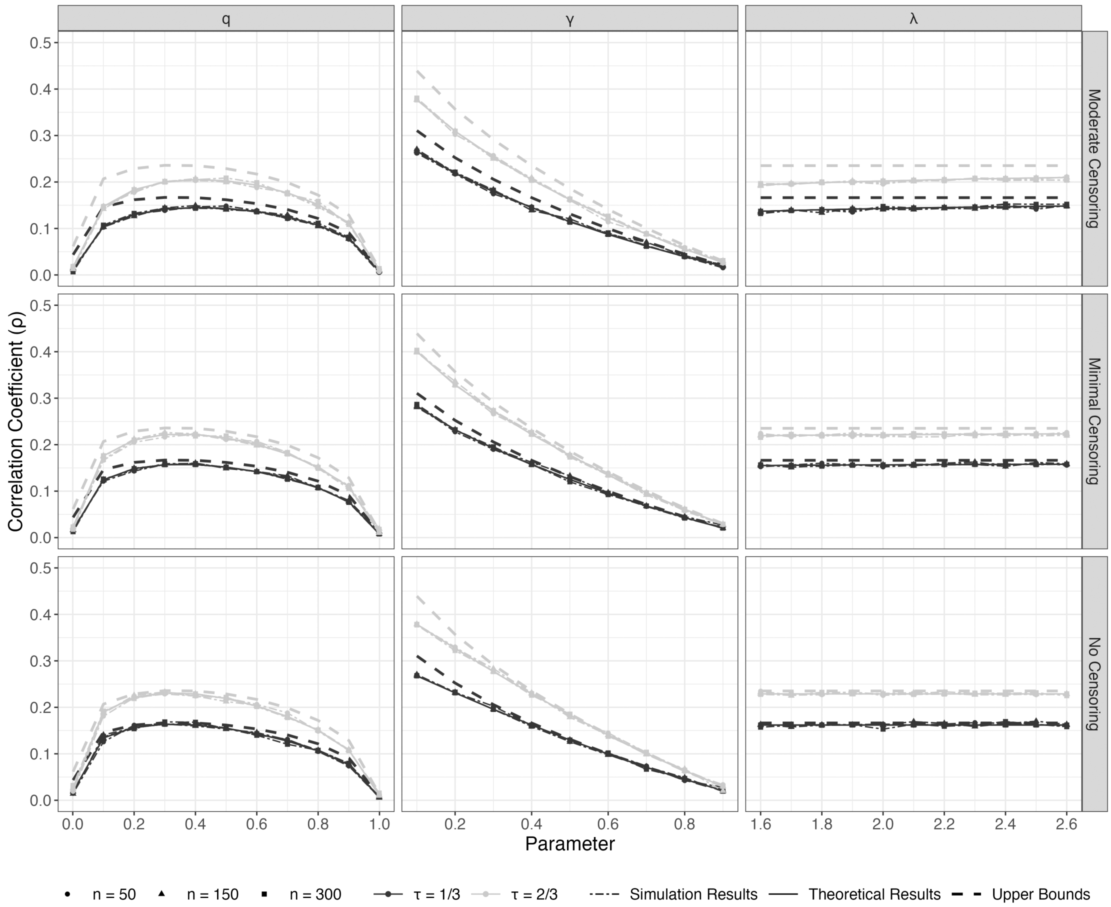


**FIGURE 3** | Relationships between the correlation coefficient $\rho$ and various factors: Response rate $q$ (left panel), PFS hazard ratio $\gamma$ between responders and non-responders (central panel), and hazard rate of non-responders $\lambda$ (right panel). For each panel of these factors, the other two factors are held constant. The constant values are $q = 0.4$, $\gamma = 0.4$, and $\lambda = 2.2$. For example, when $\gamma$ ranges from $[0.1, 0.9]$, we fix $q = 0.4$ and $\lambda = 2.2$. The upper bounds are based on Equation (5).

confidence interval of the hazard ratio between responders and non-responders was $[0.14, 0.34]$.

For simplicity, we identify $\tilde{\alpha}$ by ignoring the interim analysis and get the O'Brien–Fleming boundaries for the interim and final analysis under the identified $\tilde{\alpha}$. Following (10), let $Q_q = [0.19, 0.32]$ and $Q_\gamma = [0.14, 0.34]$, we obtain $\tilde{\alpha} = 0.018$ through a grid search. The corresponding O'Brien–Fleming boundaries are 2.985 at the interim analysis and 2.110 at the final analysis.

As a comparison, by naively assuming $\rho = 1$, the $\tilde{\alpha}$ at the final analysis will be 0.013, and the corresponding O'Brien–Fleming boundaries are 3.166 at the interim analysis and 2.239 at the final analysis.

### 4.3 | Final Design and Operating Characteristics

Given the complexity of the study design, the sample size is fine-tuned by evaluating the study operating characteristics of several scenarios (Table 2). All scenarios assume the hazard ratio of $\gamma = 0.15$ between responders and non-responders, and the PFS time of each arm follows an exponential distribution. Details on data generation are available in the Supporting Information.

**TABLE 2** | Scenarios considered for evaluating the study's operating characteristics.

| | pCR Rate ($q$) | | | mPFS in Days | | |
|---|---|---|---|---|---|---|
| **Scenario** | **SOC** | **LD** | **HD** | **SOC** | **LD** | **HD** |
| I | 0.2 | 0.2 | 0.5 | 180 | 180 | 300 |
| II | 0.2 | 0.4 | 0.5 | 180 | 276 | 300 |
| III | 0.2 | 0.5 | 0.5 | 180 | 300 | 300 |

In each scenario, starting from $D = 162$ events, we conducted a grid search to ensure that the total number of events would provide 90% power. Note that in Scenarios II and III, rejecting the null hypothesis for either HD or LD is considered a study success. Once the total number of events is determined, we set $\boldsymbol{q} = (0.2, 0.2, 0.2)$ and $\mathbf{mPFS} = (180, 180, 180)$ for the SOC, LD, and HD arms, respectively, to evaluate the FWER.

Based on 10 000 replications, Table 3 presents the total number of events required to achieve 90% power and the corresponding study operating characteristics. For comparison, we also provide the number of events required when naively assuming $\rho = 1$. Furthermore, as mentioned in the introduction, the combination test



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**TABLE 3** | Study operating characteristics corresponding to the proposed method (Proposed), assuming $\rho = 1$ ($\rho = 1$) and the combination test with the Bonferroni combination function (Combination). P(HD): Probability of selecting HD; $D$: Total number of events; Rej. Any: Rejecting $H_0$ for HD or LD; Rej. LD: Rejecting $H_0$ for LD; Rej. HD: Rejecting $H_0$ for HD. Duration: Study duration in days.

| | | | | Under $H_1$ | | | | Under $H_0$ | |
|---|---|---|---|---|---|---|---|---|---|
| **Scenario** | **P(HD)** | **Design** | $D$ | **Rej. Any** | **Rej. LD** | **Rej. HD** | **Duration** | **FWER** | **Duration** |
| I | 1.000 | Proposed | 178 | 0.901 | 0.000 | 0.901 | 373 | 0.023 | 359 |
| | | $\rho = 1$ | 192 | 0.901 | 0.000 | 0.901 | 414 | 0.020 | 391 |
| | | Combination | 197 | 0.900 | 0.000 | 0.900 | 418 | 0.016 | 404 |
| II | 0.723 | Proposed | 177 | 0.901 | 0.237 | 0.664 | 369 | 0.022 | 357 |
| | | $\rho = 1$ | 192 | 0.900 | 0.236 | 0.664 | 413 | 0.017 | 391 |
| | | Combination | 193 | 0.900 | 0.237 | 0.664 | 406 | 0.018 | 394 |
| III | 0.259 | Proposed | 162 | 0.902 | 0.663 | 0.239 | 341 | 0.020 | 326 |
| | | $\rho = 1$ | 173 | 0.903 | 0.666 | 0.237 | 368 | 0.015 | 348 |
| | | Combination | 173 | 0.905 | 0.669 | 0.237 | 359 | 0.015 | 348 |

is an alternative approach for controlling the FWER. Accordingly, we also present the results for a combination test that utilizes the Bonferroni combination function—see the technical details in the Supporting Information.

The results indicate that the proposed design controls the FWER, reduces study time by approximately 10% compared to the design using $\rho = 1$ and the design using the combination test.

Considering all the information, the finalized design requires a total of 178 events for the final PFS analysis.

## 5 | Software

We have developed and published the R package, `dtlcor`, on CRAN to implement the proposed DTL design. This package includes functions to calculate the correlation boundary $\rho^*$ and $\tilde{\alpha}$, as well as functions to conduct simulation studies to obtain design operating characteristics. Additionally, it features an R Shiny application for interactive implementation of the design. Instructions for installing and using the R package are provided in Appendix C.

## 6 | Discussion

The DTL design is gaining popularity in oncology trials as it allows for flexible exploration of multiple dose levels. In this paper, we focus on using intermediate or surrogate endpoints to determine which dose level should be dropped during the DTL analysis. Throughout the paper, we use pathological response status as an illustrative example. However, the proposed design is applicable to any intermediate binary outcomes, such as minimal residual disease status, circulating tumor DNA clearance, or tumor shrinkage status [22], provided there is existing evidence regarding their hazard ratios in relation to survival.

Instead of assuming the correlation between the test statistics of the surrogate and the primary time-to-event endpoints to be 1, which is the most conservative approach, we derive theoretical formulas for the correlation coefficient based on practically reasonable assumptions. Additionally, we derive a theoretical upper bound for the correlation that is independent of the censoring distribution and the distributions of responders and non-responders. We believe that applying this correlation upper bound for multiplicity control will allow oncology studies using DTL designs to maintain well-controlled family-wise error rates without being overly conservative or unnecessarily wasting resources.

It is noteworthy that in all the simulation scenarios we have considered, the correlation has never exceeded 0.7. While we do not have a theoretical proof for this boundary, we are confident that $\rho = 0.7$ can serve as a practical rule of thumb in the DTL design we propose.

In practice, as illustrated by the Application, once $\tilde{\alpha}$ has been determined, we recommend conducting simulation studies to fine-tune the sample size calculated by the typical approach in survival analysis. Given the design's complexity, we believe this is a necessary step.

It is also noteworthy that the DTL design considered in this paper differs from the settings considered by the alternative multi-arm multi-stage designs [23–25]. First, the design we consider does not involve hypothesis testing with respect to the surrogate endpoint. Specifically, it does not compare any treatment arm to the control based on the surrogate endpoint. The "losers" in this context are not necessarily ineffective compared to the control, but are instead less likely to be the "optimal" arm based on clinical and other considerations. Second, the proposed design focuses on identifying the most conservative estimate (i.e., the upper bound) of the correlation that is acceptable within a regulatory decision-making framework, rather than addressing the correlation between surrogate and primary endpoints from an estimation perspective. Regarding the combination test [18, 19], the design we consider does not clearly define the subset of control patients (or information) corresponding to the $p$ value of the "first



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stage," as there is no hypothesis testing at the DTL look. In the simulation conducted in Section 4.3, all control patients enrolled prior to the DTL look are included in the "first stage" and compared with the patients in the selected arm who contributed to the DTL analysis. However, alternative approaches exist (e.g., using a randomly selected subset of control subjects), which may introduce ambiguity in the study design.

In summary, the proposed design will result in a smaller sample size and shorter study duration compared to the naive and most conservative approach, which sets the correlation between the DTL statistics and the final analysis statistics to 1. We believe this will lead to more efficient oncology trials.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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### Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1.** Supporting Information.

## Appendix A

### Proofs Related to the Type I Error Inflation

In Section 2.2, Example 1, $(Z_{10}, Z_{20}, Z_{11}, Z_{21})$ follows a multivariate normal distribution with mean 0 and variance $\mathbf{\Sigma}$.

Then, we have

$$\begin{bmatrix} Z_{20} - Z_{10} \\ Z_{11} \end{bmatrix} \sim N\left(\mathbf{0}, \begin{bmatrix} 1 & -\frac{1}{2}\sqrt{\frac{t_0}{t_1}} \\ -\frac{1}{2}\sqrt{\frac{t_0}{t_1}} & 1 \end{bmatrix}\right)$$

$$\begin{bmatrix} Z_{20} - Z_{10} \\ Z_{21} \end{bmatrix} \sim N\left(\mathbf{0}, \begin{bmatrix} 1 & \frac{1}{2}\sqrt{\frac{t_0}{t_1}} \\ \frac{1}{2}\sqrt{\frac{t_0}{t_1}} & 1 \end{bmatrix}\right)$$



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Subsequently, the conditional distributions of $(Z_{20} - Z_{10})|Z_{11}$ and $(Z_{20} - Z_{10})|Z_{21}$ are

$$(Z_{20} - Z_{10})|Z_{11} \sim N\left(-\frac{1}{2}\sqrt{\frac{t_0}{t_1}}Z_{11}, 1 - \frac{t_0}{4t_1}\right)$$
$$(Z_{20} - Z_{10})|Z_{21} \sim N\left(\frac{1}{2}\sqrt{\frac{t_0}{t_1}}Z_{21}, 1 - \frac{t_0}{4t_1}\right)$$

Then, the FWER is given by:

$$\begin{aligned} r &= \Pr(Z_{20} - Z_{10} - \Delta \leq 0, Z_{11} > z_{\alpha^*}) + \Pr(Z_{20} - Z_{10} - \Delta \\ &\quad > 0, Z_{21} > z_{\alpha^*}) \\ &= \Pr(Z_{20} - Z_{10} - \Delta \leq 0, Z_{11} > z_{\alpha^*}) + \Pr(Z_{21} > z_{\alpha^*}) \\ &\quad - \Pr(Z_{20} - Z_{10} - \Delta \leq 0, Z_{21} > z_{\alpha^*}) \\ &= \alpha^* + \int_{z_{\alpha^*}}^{\infty} \Pr(Z_{20} - Z_{10} - \Delta \leq 0 | Z_{11} = z)\phi(z)dz \\ &\quad - \int_{z_{\alpha^*}}^{\infty} \Pr(Z_{20} - Z_{10} - \Delta \leq 0 | Z_{21} = z)\phi(z)dz \\ &= \alpha^* + \int_{z_{\alpha^*}}^{\infty} \left[\Phi(d_1) - \Phi(d_2)\right]\phi(z)dz \end{aligned}$$

where $z_{\alpha^*}$ is the $(1 - \alpha^*)$ percentile of the standard normal distribution, $\phi(\cdot)$ and $\Phi(\cdot)$ represent the standard normal PDF and CDF, respectively, and

$$d_1 = \frac{2\Delta + z\sqrt{t_0/t_1}}{\sqrt{4 - t_0/t_1}}, \ d_2 = \frac{2\Delta - z\sqrt{t_0/t_1}}{\sqrt{4 - t_0/t_1}}$$

In Example 2, we have

$$\begin{bmatrix} W_2 - W_1 \\ Z_{11} \end{bmatrix} \sim N\left(\mathbf{0}, \begin{bmatrix} \frac{2q(1-q)}{n} & -\rho\sqrt{\frac{q(1-q)}{n}} \\ -\rho\sqrt{\frac{q(1-q)}{n}} & 1 \end{bmatrix}\right)$$
$$\begin{bmatrix} W_2 - W_1 \\ Z_{21} \end{bmatrix} \sim N\left(\mathbf{0}, \begin{bmatrix} \frac{2q(1-q)}{n} & \rho\sqrt{\frac{q(1-q)}{n}} \\ \rho\sqrt{\frac{q(1-q)}{n}} & 1 \end{bmatrix}\right)$$

The conditional distributions of $(W_2 - W_1)|Z_{11}$ and $(W_2 - W_1)|Z_{21}$ are

$$(W_2 - W_1)|Z_{11} \sim N\left(-\rho Z_{11}\sqrt{q(1-q)/n}, (2 - \rho^2)q(1-q)/n\right)$$
$$(W_2 - W_1)|Z_{21} \sim N\left(\rho Z_{21}\sqrt{q(1-q)/n}, (2 - \rho^2)q(1-q)/n\right)$$

Then, the FWER becomes

$$\begin{aligned} r &= \Pr(W_2 - W_1 - \Delta \leq 0, Z_{11} > z_{\alpha^*}) \\ &\quad + \Pr(W_2 - W_1 - \Delta > 0, Z_{21} > z_{\alpha^*}) \\ &= \Pr(W_2 - W_1 - \Delta \leq 0, Z_{11} > z_{\alpha^*}) + \Pr(Z_{21} > z_{\alpha^*}) \\ &\quad - \Pr(W_2 - W_1 - \Delta \leq 0, Z_{21} > z_{\alpha^*}) \\ &= \alpha^* + \int_{z_{\alpha^*}}^{\infty} \Pr(W_2 - W_1 - \Delta \leq 0 | Z_{11} = z)\phi(z)dz \\ &\quad - \int_{z_{\alpha^*}}^{\infty} \Pr(W_2 - W_1 - \Delta \leq 0 | Z_{21} = z)\phi(z)dz \\ &= \alpha^* + \int_{z_{\alpha^*}}^{\infty} \left[\Phi(d_1) - \Phi(d_2)\right]\phi(z)dz \end{aligned}$$

where

$$d_1 = \frac{\Delta/\sqrt{q(1-q)/n} + \rho z}{\sqrt{2 - \rho^2}}, \ d_2 = \frac{\Delta/\sqrt{q(1-q)/n} - \rho z}{\sqrt{2 - \rho^2}}$$

## Appendix B

### Proofs Related to the Correlation Coefficient $\rho$

#### Proof of Propositions

From Section 2.3, under the global null hypothesis, the correlation coefficient, $\mathrm{Cor}(W_j, Z_{jk})$ can be derived as follows:

$$\begin{aligned} \mathrm{Cor}(W_j, Z_{jk}) = \rho_{jk} &= \frac{\mathrm{Cov}\left(\sum_{i=1}^n X_{ij}/n, Z_{jk}\right)}{\sqrt{q(1-q)/n}} = \frac{\sum_{i=1}^n E(X_{ij}Z_{jk})}{\sqrt{nq(1-q)}} \\ &= \frac{\sum_{i=1}^n E(E(X_{ij}Z_{jk}|X_{ij}))}{\sqrt{nq(1-q)}} \\ &= \frac{\sum_{i=1}^n E(Z_{jk}|X_{ij} = 1)\Pr(X_{ij} = 1)}{\sqrt{nq(1-q)}} \\ &= \sqrt{\frac{nq}{1-q}}E(Z_{jk}|X_{1j} = 1) \end{aligned}$$

The process of calculating $E(Z_{jk}|X_{1j} = 1)$ is outlined below.

First, the conditional distribution, survival function, and hazard function for $T_{ij}$ given $X_{1j} = 1$ in treatment arm $j$ ($j = 1, \ldots, J$) are as follows:

$$\begin{aligned} f_k^*(t) &= \frac{1}{n_k}f_1(t) + \frac{n_k - 1}{n_k}f(t) \\ S_k^*(t) &= \frac{1}{n_k}S_1(t) + \frac{n_k - 1}{n_k}S(t) \\ h_k^*(t) &= f_k^*(t)/S_k^*(t) \end{aligned}$$

Then, according to [26] and [27], the distribution of $Z_{jk}$ given $X_{1j} = 1$ follows $N(\psi_{jk}, 1)$ with

$$\psi_{jk} = \frac{\sqrt{2n_k}\int_0^\infty \log\left(\frac{h(t)}{h_k^*(t)}\right)\pi_k(t)(1 - \pi_k(t))V_k(t)dt}{\sqrt{\int_0^\infty \pi_k(t)(1 - \pi_k(t))V_k(t)dt}}$$

with

$$\begin{aligned} V_k(t) &= \frac{1}{2}\left[f(t) + f_k^*(t)\right]S_C(t) \\ \pi_k(t) &= \frac{S_k^*(t)}{S(t) + S_k^*(t)} \end{aligned}$$

Subsequently, the correlation coefficient can be written as

$$\rho_{jk} = \frac{\sqrt{\frac{2qnn_k}{1-q}}\int_0^\infty \log\left(\frac{h(t)}{h_k^*(t)}\right)\pi_k(t)(1 - \pi_k(t))V_k(t)dt}{\sqrt{\int_0^\infty \pi_k(t)(1 - \pi_k(t))V_k(t)dt}} \tag{B1}$$

In the following, (B1) is further simplified.

When $n_k$ is large, $S_k^*(t) \to S(t)$ and $f_k^*(t) \to f(t)$. Subsequently, $\pi_k(t) \to 1/2$ and $V(t) \to f(t)S_C(t)$. Thus, we can approximate $\rho_{jk}$ in Equation (B1) by

$$\rho_{jk} \approx \frac{\sqrt{\frac{qnn_k}{2(1-q)}}\int_0^\infty \log\frac{h(t)}{h_k^*(t)}f(t)S_C(t)dt}{\sqrt{\int_0^\infty f(t)S_C(t)dt}} \tag{B2}$$



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Furthermore, note that

$$\log\left(\frac{h(t)}{h_k^*(t)}\right) = \log\left(\frac{S_k^*(t)}{S(t)}\right) - \log\left(\frac{f_k^*(t)}{f(t)}\right)$$

and $S_k^*(t)$ and $f_k^*(t)$ are close to $S(t)$ and $f(t)$ when $n_k$ is large, respectively, $\log(h(t)/h_k^*(t))$ can be approximated by using the first-order Taylor expansion for $\log(f_k^*(t)/f(t))$ and $\log(S_k^*(t)/S(t))$ with respect to $f_k^*(t)/f(t)$ and $S_k^*(t)/S(t)$. Then, the integral part of the numerator of (B2) becomes

$$\begin{aligned}
&\int_0^\infty \log\left(\frac{h(t)}{h_k^*(t)}\right) f(t)S_C(t)dt \\
&\approx \int_0^\infty \left[\frac{S_k^*(t)}{S(t)} - \frac{f_k^*(t)}{f(t)}\right] f(t)S_C(t)dt \\
&= \int_0^\infty \left[\frac{S_k^*(t)f(t)}{S(t)} - f_k^*(t)\right] S_C(t)dt \\
&= \frac{1}{n_k}\int_0^\infty \left[\frac{S_1(t)f(t)}{S(t)} - f_1(t)\right] S_C(t)dt
\end{aligned}$$

Let $\tau_k = n/n_k$. Then, Equation (B2) becomes

$$\rho_{jk} \approx \frac{\sqrt{\frac{q\tau_k}{2(1-q)}}\int_0^\infty \left[\frac{S_1(t)f(t)}{S(t)} - f_1(t)\right] S_C(t)dt}{\sqrt{\int_0^\infty f(t)S_C(t)dt}} \tag{B3}$$

In a proportional hazards situation, denote $\gamma$ the hazard ratio between subjects with pCR vs. those without, $\gamma > 0$. Let $h_0(t)$ and $h_1(t)$ be the hazard functions of $f_0(t)$ and $f_1(t)$, respectively. Then, $h_1(t) = h_0(t)\gamma$, $S_1(t) = [S_0(t)]^\gamma$, the integral term of the numerator of Equation (B3) becomes

$$\begin{aligned}
&\int_0^\infty \left[\frac{S_1(t)f(t)}{S(t)} - f_1(t)\right] S_C(t)dt \\
&= \int_0^\infty \left[\frac{qS_1(t)h_1(t) + (1-q)S_0(t)h_1(t)/\gamma}{qS_1(t) + (1-q)S_0(t)} S_1(t) - f_1(t)\right] S_C(t)dt \\
&= \int_0^\infty \left[\frac{1 + \frac{1-q}{q}[S_1(t)]^{\frac{1}{\gamma}-1}/\gamma}{1 + \frac{1-q}{q}[S_1(t)]^{\frac{1}{\gamma}-1}} f_1(t) - f_1(t)\right] S_C(t)dt \\
&= \int_0^\infty \frac{\left(\frac{1}{\gamma}-1\right)\frac{1-q}{q}[S_1(t)]^{\frac{1}{\gamma}-1}}{1 + \frac{1-q}{q}[S_1(t)]^{\frac{1}{\gamma}-1}} f_1(t)S_C(t)dt \\
&= \left(\frac{1}{\gamma}-1\right)\int_0^\infty \frac{1}{1 + \frac{q}{1-q}[S_1(t)]^{1-\frac{1}{\gamma}}} f_1(t)S_C(t)dt
\end{aligned} \tag{B4}$$

Thus, Equation (B3) becomes

$$\rho_{jk} \approx \frac{\sqrt{\frac{q\tau_k}{2(1-q)}}\left(\frac{1}{\gamma}-1\right)\int_0^\infty \frac{1}{1+\frac{q}{1-q}[S_1(t)]^{1-\frac{1}{\gamma}}} f_1(t)S_C(t)dt}{\sqrt{\int_0^\infty f(t)S_C(t)dt}} \tag{B5}$$

Considering the case without censoring, Equation (B3) can be further reduced to

$$\rho_{jk} \approx \sqrt{\frac{q\tau_k}{2(1-q)}}\left[\int_0^\infty \frac{S_1(t)f(t)}{S(t)}dt - 1\right] \tag{B6}$$

Under the proportional hazard model for $T_j|X_j$, Equation (B4) becomes

$$\int_0^\infty \frac{S_1(t)f(t)}{S(t)}dt - 1 = \left(\frac{1}{\gamma}-1\right)E\left[\frac{1}{1+\frac{q}{1-q}U^{1-\frac{1}{\gamma}}}\right]$$

where $U \sim \text{Unif}(0,1)$. Thus, Equation (B6) becomes

$$\rho_{jk} \approx \sqrt{\frac{q\tau_k}{2(1-q)}}\left(\frac{1}{\gamma}-1\right)E\left[\frac{1}{1+\frac{q}{1-q}U^{1-\frac{1}{\gamma}}}\right]$$

**Proof of Theorem**

By Hölder's inequality,

$$\begin{aligned}
&\int_0^\infty \frac{1}{1+\frac{q}{1-q}[S_1(t)]^{1-\frac{1}{\gamma}}} f_1(t)S_C(t)dt \\
&= \int_0^\infty \frac{1}{\left(1+\frac{q}{1-q}[S_1(t)]^{1-\frac{1}{\gamma}}\right)\left(q+\frac{1-q}{\gamma}[S_1(t)]^{\frac{1}{\gamma}-1}\right)} f(t)S_C(t)dt \\
&\leq \sqrt{\int_0^\infty \frac{1}{\left(1+\frac{q}{1-q}[S_1(t)]^{1-\frac{1}{\gamma}}\right)^2\left(q+\frac{1-q}{\gamma}[S_1(t)]^{\frac{1}{\gamma}-1}\right)^2} f(t)S_C(t)dt \times \int_0^\infty f(t)S_C(t)dt}
\end{aligned}$$

Hence, for $\gamma \leq 1$, Equation (B5) can be bounded by

$$\begin{aligned}
\rho_{jk} &\leq \sqrt{\frac{q\tau_k}{2(1-q)}}\left(\frac{1}{\gamma}-1\right)\sqrt{\int_0^\infty \frac{1}{\left(1+\frac{q}{1-q}[S_1(t)]^{1-\frac{1}{\gamma}}\right)^2\left(q+\frac{1-q}{\gamma}[S_1(t)]^{\frac{1}{\gamma}-1}\right)^2} f(t)S_C(t)dt} \\
&\leq \sqrt{\frac{q\tau_k}{2(1-q)}}\left(\frac{1}{\gamma}-1\right)\sqrt{\int_0^\infty \frac{1}{\left(1+\frac{q}{1-q}[S_1(t)]^{1-\frac{1}{\gamma}}\right)^2\left(q+\frac{1-q}{\gamma}[S_1(t)]^{\frac{1}{\gamma}-1}\right)^2} f_1(t)dt} \\
&= \sqrt{\frac{q\tau_k}{2(1-q)}}\left(\frac{1}{\gamma}-1\right)\sqrt{E\left[\frac{1}{\left(1+\frac{q}{1-q}U^{1-\frac{1}{\gamma}}\right)^2\left(q+\frac{1-q}{\gamma}U^{\frac{1}{\gamma}-1}\right)}\right]}
\end{aligned} \tag{B7}$$

where $U \sim \text{Unif}(0,1)$.



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## Appendix C

### Example of Using the Software

The following code demonstrates how to install the R package `dtlcor` and use it to calculate $\widetilde{\alpha}$ and conduct simulation studies of the proposed design.

```
## install dtlcor package
install.packages(``dtlcor'')
require(dtlcor)

## get tilde alpha
alpha_t <- dtl_app_get_alpha_t(n = 80, N = 152,
 q_seq = seq(0.19, 0.32, 0.01), gamma_seq = seq(0.14, 0.34, 0.01),
 alpha = 0.025, delta = 0.05)

## simulate a single trial
trial <- dtl_app_sim_single(D = 162, n = 80, N = 152,
 mPFS = c(180, 276, 300), q = c(0.2, 0.4, 0.5), gamma = 0.15,
 drop_rate = 0.05, enroll = 20 * 12, interim_t = c(0.5, 1), delta = 0.05)
```



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